

Model Scope

Outputs wanted:

- Optimal fleet sizes: N_{amb} , N_{Mini} (and optional new minibus variants)
- Hourly staffing curves: Drivers + Vehicle-agents + pushers: $H_{Drv,t}$, $H_{VehAg,t}$, $H_{Push,t}$
- Mode decisions (Scenario 2-3): Ambulift vs minibus vs pusher for horizontal legs
- SLA performance: arrivals < 20 mins, departures < 30 mins, aiming 99-100%

1. Sets and Indices

Time/Uncertainty

- $t \in T$ time buckets (e.g. 15-min or 60-min)
 $\omega \in \Omega$ stochastic scenarios (delays, congestion, downtime)

Demand entities

- $j \in J$ PRM jobs / passengers needing service
 $f \in F$ flights
 $J_f \subseteq J$ jobs belonging to flight f

Classification sets

- $d \in D = \{Arr, Dep\}$ direction
 $c \in C = \{Dom, Int\}$ flight class (domestic/international)
 $z \in Z$ stand-zones for batching
 $s \in S$ stands
 $g \in G = \{7, 8\}$ lift-gates group (Gate 7/8)

Resource/mode sets

- $v \in V = \{Ambulift, Minibus\}$ vehicle types
 $m \in M = \{Amb, Mini, Push\}$ horizontal modes
 $r \in R = \{Drv, VehAg, Push\}$ staff roles

2. Parameters (inputs)

Job Attributes

For each job j :

- $dir_j \in D$
 $flight(j) = f \in F$
 $stand_j \in S$, $zone_j \in Z$
 $class_f \in C$ (Dom/Int)
 $k_f \in \{0,1\}$ needs wheelchair slot (WCHC or WCHS own chair)
 $m_j \subseteq M$ allowed horizontal modes for this job
 $vert_j \in \{0,1\}$ Vertical Component Required
 $t(j) \in T$ time bucket in which the job is released/needs service

SLA Limits

$$L_j = \begin{cases} 20 & \text{if } dir_j = Arr \\ 30 & \text{if } dir_j = Dep \end{cases}$$

Costs

$$c_v^{capex} \text{ (£/hour)}$$
$$c_r^{wage} \text{ (£/hour)}$$

Vehicle capacities

Per vehicle type v :

$seatcap_v, wccap_v$

Workload / time consumption coefficients

for each job j , mode m , scenario w , and time bucket t we have “minutes consumed” parameters:

a) Vehicle busy minutes

$$\tau_{j,v,\omega}^{veh}(\text{minutes}) \text{ for } v \in \{Ambulift, Minibus\}$$

b) Staff minutes

$$\tau_{j,v,\omega}^{drv}, \tau_{j,v,\omega}^{vag}, \tau_{j,\omega}^{push}$$

Spin locking

Let:

j^A = arrival PRM job for a flight

j^D = departure PRM job for the same aircraft

$spin_f = \begin{cases} 1 & \text{if arrival and departure are within spin window and require vertical} \\ 0 & \text{otherwise} \end{cases}$

$z_f = \begin{cases} 1 & \text{if ambulift is pinned to flight } f \\ 0 & \text{otherwise} \end{cases}$

For each bucket t, ω :

$$k_{t,\omega}^{spin} \quad (\text{Ambulift-minutes unavailable due to spins/lock on rules})$$

Gate 7/8

Let $F_{78} \subseteq F$ be flights that require G 7/8 lift flow and let w_f be the number of **WCH** PRMs on flight f

g (lift minutes per wheelchair cycle),

L_t (lift minutes available in bucket t)

3. Decision Variables

Fleet sizing

Number of vehicles of each type available in PRM fleet

$$N_v \in \mathbb{Z}_{\geq 0} \quad \forall v \in V$$

Hourly Staffing

Number of staff of role r available in time period t

$$H_{r,t} \in \mathbb{Z}_{\geq 0} \quad \forall r \in R, t \in T$$

Horizontal mode choice (Scenario 2-3)

$$x_{j,m,\omega} \in \{0,1\} \quad \forall j \in J, m \in M, \omega \in \Omega$$

$x_{j,m,\omega} = 1$ means job j uses horizontal mode m under scenario ω .

Scenario 1 will fix these using decision-tree rules

SLA violation indicator

Binary variable equal to 1 if PRM jobs breaches the ECAC SLA in scenario ω

$$y_{j,\omega} \in \{0,1\} \quad \forall j, \omega$$

Batching variables (Scenario 3 only)

Variable representing how many minibus trips are created per (zone, direction, class, time):

$$k_{z,d,c,t,\omega} \in \mathbb{Z}_{\geq 0} \quad \forall z, d, c, t, \omega$$

Variable representing how many PRMs from each flight are assigned to minibus batching:

$$u_{f,t,\omega} \in \mathbb{Z}_{\geq 0} \quad \forall f, t, \omega \quad \text{where } u_{f,t,\omega} \text{ is the number of PRMs from flight } f \text{ served by minibus}$$

in bucket time

Wheelchair slot tracking:

$$u_{f,t,\omega}^{wch} \in \mathbb{Z}_{\geq 0}$$

4. Objective Function

$$\min \sum_{v \in V} c_v^{capex} N_v |T| + \sum_{t \in T} \sum_{r \in R} c_r^{wage} H_{r,t} + \lambda \sum_{\omega \in \Omega} \sum_{j \in J} y_{j,\omega}$$

λ chosen large so SLA breaches are expensive

5. Constraints

- a) Each job chooses exactly one horizontal mode (Scenario 2-3)

$$\sum_{m \in M} x_{j,m,\omega} = 1 \quad \forall j, \omega$$

Note scenario 1: set $x_{j,m,\omega} = \bar{x}_{j,m}$ from decision tree

- b) Mode eligibility constraints (Scenario 2-3)

Maintain a set $m_j \subseteq M$ of allowed horizontal modes per job j . Then:

$$x_{j,m,\omega} = 0 \quad \forall j, \forall m \notin M_j, \forall \omega$$

Encode rules such as:

- Safety stands (15, 15A, 15B and 17) disallow pusher only when the flight is not ryanair because no safe walking route
- Domestic arrival allows Ambulift horizontally

- c) Vehicle time availability

Ambulifts

$$\sum_{j:t(j)=t} \tau_{j,Amb,\omega}^{veh} x_{j,Amb,\omega} \leq 60 N_{Amb} - k_{t,\omega}^{spin} \quad \forall t, \omega$$

Minibuses

$$\sum_{j:t(j)=t} \tau_{j,Mini,\omega}^{veh} x_{j,Mini,\omega} \leq 60 N_{Mini} \quad \forall t, \omega$$

- d) Staff time availability

Drivers

$$\sum_{j:t(j)=t} (\tau_{j,Amb,\omega}^{drv} x_{j,Amb,\omega} + \tau_{j,Mini,\omega}^{drv} x_{j,Mini,\omega}) \leq 60 H_{Drv,t} \quad \forall t, \omega$$

Vehicle Agents

$$\sum_{j:t(j)=t} (\tau_{j,Amb,\omega}^{vag} x_{j,Amb,\omega} + \tau_{j,Mini,\omega}^{vag} x_{j,Mini,\omega}) \leq 60 H_{VehAg,t} \quad \forall t, \omega$$

Pushers

$$\sum_{j:t(j)=t} \tau_{j,\omega}^{push} x_{j,Push,\omega} \leq 60 H_{Push,t} \quad \forall t, \omega$$

- e) Vehicle Physical Capacity

Let:

$k_j = 1$ if passenger needs wheelchair space, 0 otherwise

Wheelchair slot capacity (vehicle)

$$\sum_{j:t(j)=t} k_j x_{j,v,\omega} \leq wccap_v \cdot N_v \quad \forall t, \omega$$

f) Gate 7/8 bottleneck

$z_f = 1$: flight f uses minibus strategy for WCH PRMs

$z_f = 0$: uses lift/pusher flow (consumes lift capacity)

Lift time capacity constraint:

$$\sum_{f \in F_{78t}(f)=t} gw_f(1 - z_f) \leq L_t \quad \forall t$$

g) SLA breach definition (all scenarios)

Let $R_{j,\omega}$ = response time minutes for job j in scenario ω

$$R_{j,\omega} \leq L_j + M y_{j,\omega} \quad \forall j, \omega$$

Enforce 99-100% hard limit:

$$\frac{1}{|J||\Omega|} \sum_{\omega \in \Omega} \sum_{j \in J} y_{j,\omega} \leq 0.01$$

ADD IN SLA PENALTIES RELATED TO DIFFERENT AREAS ??

h) Batching constraints (scenario 3 only)

Demand assignment

Let $n_{f,t}$ be number of PRMs on flight f in time bucket t

$$0 \leq u_{f,t,\omega} \leq n_{f,t} \quad \forall f, t, \omega$$

Only batch within same zone, direction, class

For each batch key (z, d, c, t) , total PMS served by minibuses in that bucket must fit in trips

$$\sum_{f: zone_f=z, dir_f=d, class_f=c} u_{f,t,\omega} \leq seatcap_v \cdot k_{v,z,d,c,t,\omega} \quad \forall z, d, c, t, \omega$$

Link batching to job-level assignment

$$\sum_{j \in J: t(j)=t} x_{j,v,\omega} = u_{f,t,\omega} \quad \forall f, t, \omega$$

Vehicle time availability depends on number of trips

Let $\tau_{v,z,d,c,t,\omega}^{trip}$ = expected minutes a vehicle is tied up per batch trip for that flow/zone

$$\sum_{z,d,c} \tau_{z,d,c,t,\omega}^{trip} k_{v,z,d,c,t,\omega} \leq 60N_v \quad \forall t, \omega$$

Gate 7/8 lift relief via batching

$$z_f = 1 \text{ if } \sum_t u_{f,t,\omega} \geq 1$$

Scenario Definitions

1. Current operation

- Horizontal assignments fixed using existing decision-tree logic

2. Optimised horizontal (no batching)

- Optimises horizontal mode per PRM subject to constraints

3. Optimised with batching

- Allows PRMs from compatible flights to be grouped into minibus/Ambulift trips, subject to:
 - Stand-zone rules
 - Domestic/international separation
 - Vehicle and staff capacity

